

Electrical Vehicles Analytics

Course: MIS 311- Introduction to Business Analytics

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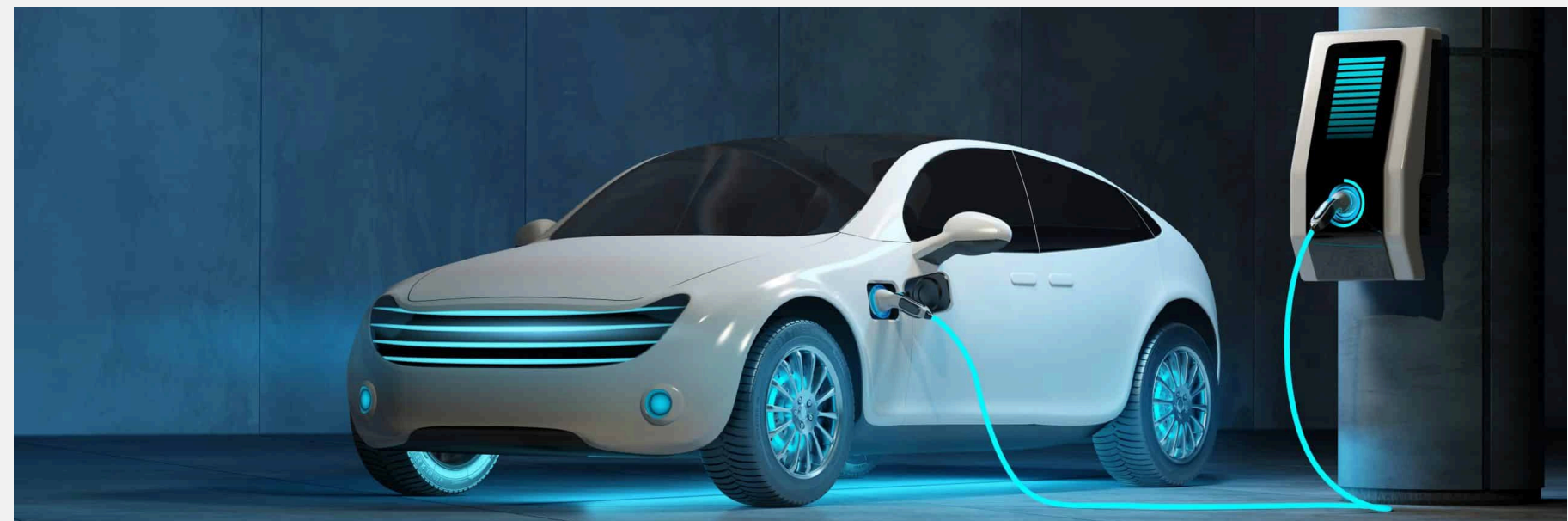
Quarter: 3

Year: 2025-2026



Project Overview

- The project analyzes electric vehicle (EV) data using Python and Power BI.
- The objective is to identify trends and patterns in the EV market.
 - The analysis focuses on:
 - Vehicle manufacturers
 - Vehicle models
 - Model years
 - Geographic distribution



Dataset Summary

Dataset Summary

- Total rows: 202
- Total columns: 6
- Dataset includes:
 - Vehicle Identification Number (VIN)
 - City
 - Postal Code
 - Model Year
 - Make
 - Model

Data Issues Found

- Missing values
- Inconsistent model names
- Invalid categorical values

id	vehicle_identification_number	city	postal_code	model_year	make
0	JTMAB3FV3P	Seabeck	98380	2023	TOYOTA
1	1N4AZ1CP6J	Bremerton	98312	2018	NISSAN
2	5YJ3E1EA4L	Seattle	98101	2020	TESLA
3	1N4AZ0CP8E	Seattle	98125	2014	NISSAN
4	5YJYGDEE5L	Lynnwood	98036	2020	TESLA
...	...	...	...	...	...
197	5LMYJ8XY4M	Snohomish	98296	2021	LINCOLN
198	1C4RJXN6XR	Pullman	99163	2024	JEEP
199	KNDPYDDH8R	Poulsbo	98370	2024	KIA
200	3MW39FS08P	Ephrata	98823	2023	BMW
201	1FAHP3R47C	Kingston	98346	2012	FORD

Data Cleaning Process

Data Cleaning Using Python

Imported and explore the dataset



01

- Used `pd.read_csv()` to import the dataset into Python for analysis.
- Used `df.info()` to examine the dataset structure, including rows, columns, and data types.
- Used `df.describe()` to generate descriptive statistics such as mean, minimum, and maximum values.

Handle Missing and Inconsistent Data



02

- Used `df.isnull().sum()` to identify and count missing values in each column.
- Used `fillna()` with the median value to replace missing numerical values in the Model Year column.
- Used `fillna()` with the mode or "Unknown" to handle missing categorical values.
- Used `replace()` to clean invalid or inconsistent values in the Model column.

Analyze and Export the Cleaned Dataset



03

- Used `value_counts()` to identify the most common vehicle manufacturers and models.
- Used `df.to_csv()` to export the cleaned dataset for Power BI visualization and dashboard creation.

Exploratory Data Analysis

1 to 11 of 11 entries Filter ?

index	vehicle_identification_number	city	postal_code	model_year	make	model
count	202	202	202.0	202.0	202	202
unique	170	52	NaN	NaN	23	44
top	Unknown	Seattle	NaN	NaN	TESLA	MODEL 3
freq	4	25	NaN	NaN	79	45
mean	NaN	NaN	98357.70792079208	2018.5891089108911	NaN	NaN
std	NaN	NaN	327.30103726031064	3.275793959894521	NaN	NaN
min	NaN	NaN	98008.0	2011.0	NaN	NaN
25%	NaN	NaN	98102.25	2017.0	NaN	NaN
50%	NaN	NaN	98290.0	2019.0	NaN	NaN
75%	NaN	NaN	98502.0	2020.0	NaN	NaN
max	NaN	NaN	99324.0	2025.0	NaN	NaN

Descriptive Statistics Findings

- Tesla was the most common manufacturer
- Model 3 was the most common vehicle model
- Average model year $\approx$ 2018.6
- Most vehicles were produced between 2018 and 2020

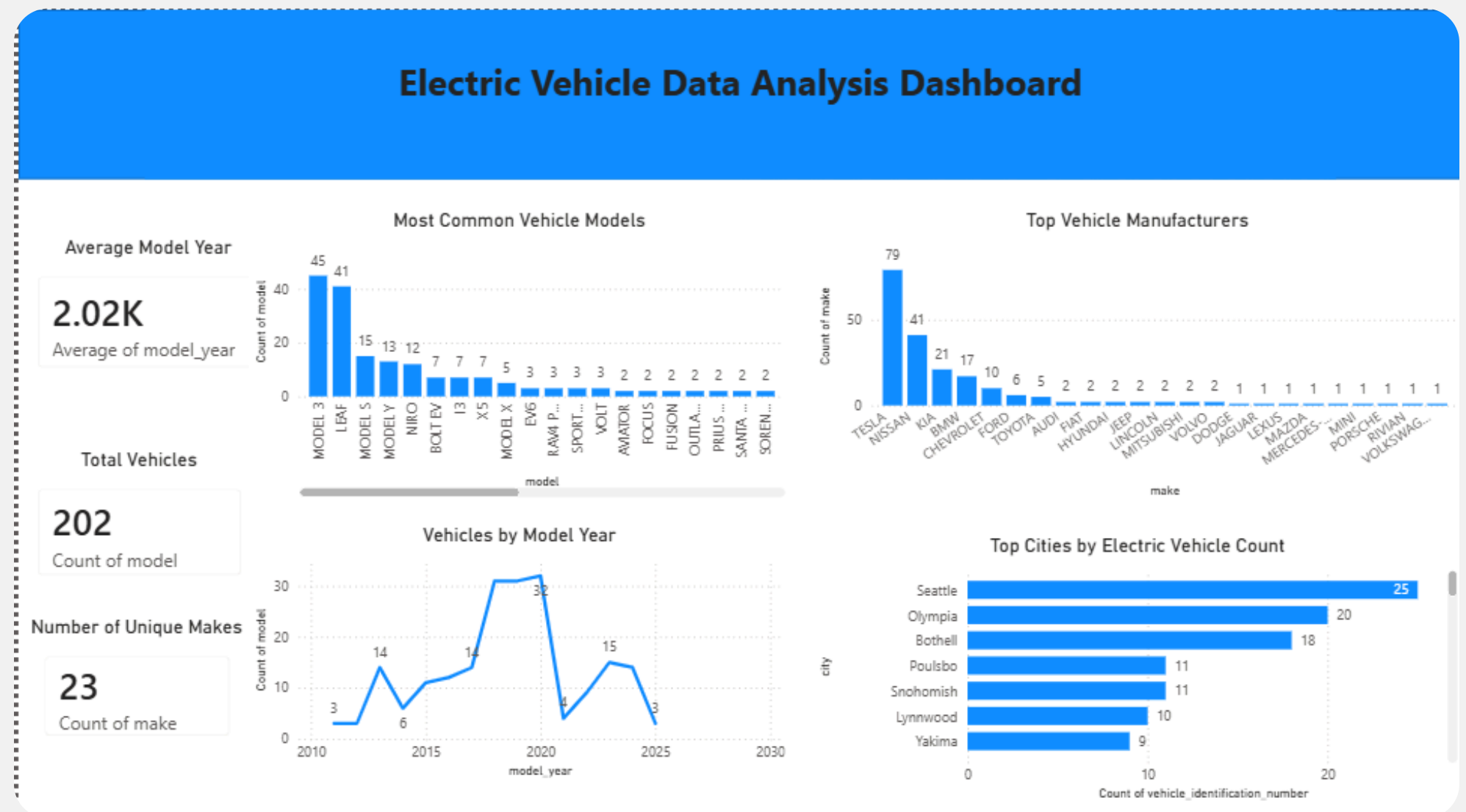
Dashboard Overview

Dashboard Components

- KPI Cards
- Bar Charts
- Line Charts
- Vehicle Distribution Analysis

Purpose

- Visualize EV market trends
- Support data-driven decision making



Key Insight 1

Finding

- Tesla is the most dominant vehicle manufacturer in the dataset, while Model 3 is the most frequently occurring vehicle model.

Implication

- Consumers strongly prefer advanced EV technology and trusted EV brands.

Business Recommendation

- Other manufacturers should invest more in innovation and smart EV technologies to remain competitive.

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Key Insight

Finding

- Most electric vehicles in the dataset were concentrated between 2018 and 2020, indicating rapid EV market growth during this period.

Implication

- Consumer demand for modern electric vehicles has increased significantly.

Business Recommendation

- Businesses and governments should continue investing in EV infrastructure and sustainable transportation technologies.

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Conclusion

- The EV market has experienced rapid growth in recent years.
- Tesla currently dominates the EV market in this dataset.
- Data analytics and visualization help businesses understand market trends and consumer behavior.
- Power BI dashboards improve business decision-making through interactive insights.



Thank You

Questions & Discussion
Thank you for listening!